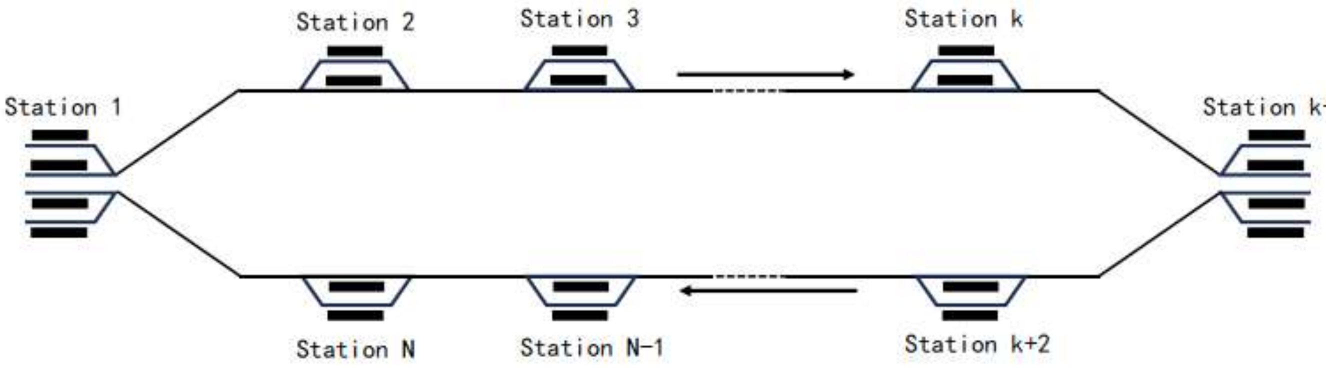
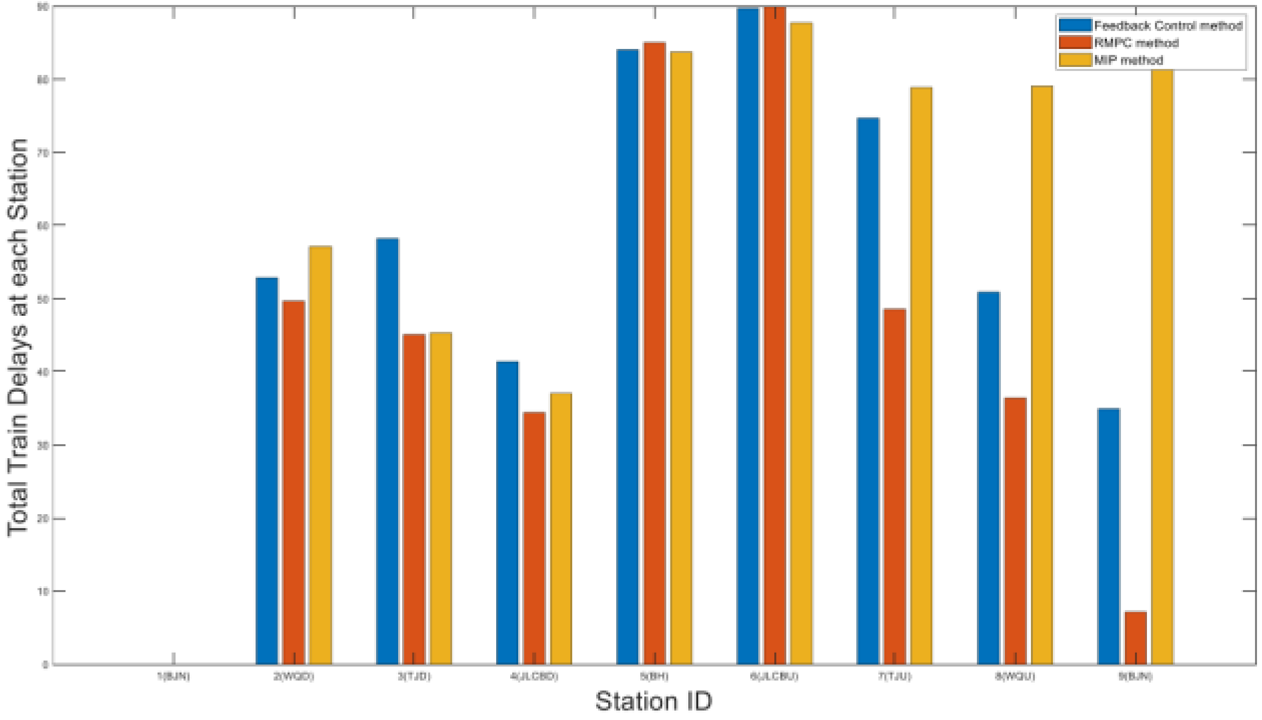
Structure of the intercity high-speed railway line



Train delay time evolution diagram of the feedback controller, the robust MPC and the MIP method



Intercity high speed railway model

State, control, and interference vectors

$$X_j^i = [\underbrace{x_j^{i-M} x_j^{i-M+1} \cdots x_j^{i-N-1}}_{M-N} \underbrace{x_j^{i-N} x_{j-1}^{i-N+1} \cdots x_{j-N+1}^{i-1}}_{N}]^T$$

$$\mathrm{U}_j^i = [\underbrace{u_{j-1}^{i-N+1} u_{j-2}^{i-N+2} \cdots u_{j-N}^{i}}_{N}]^T$$

$$W_j^i = [\underbrace{w_{j-1}^{i-N+1} w_{j-2}^{i-N+2} \cdots w_{j-N}^{i}}_{N}]^T$$

State space model

$$\begin{aligned} X_j^{i+1} =& AX_j^i + I_1 E_{1,j} C_1 X_j^i + I_1 E_{2,j} C_2 X_j^i + BU_j^i \\ &+ LE_{3,i} C_3 U_j^i + BW_j^i, \quad j \in \{1, 2, \ldots, N\} \end{aligned}$$

Robust model predictive control design

Robust MPC problem

$$\begin{aligned} &\min_{U(i+j|i)=K(i)X(i+j|i), j=0,1,2,\ldots,i_f-i} G(i) \\ &= \sum_{i=0}^{j_f-j_0} \{X(i+j \mid i)^T Q X(i+j \mid i) \\ &+ \bar{U}^T(i+j \mid i) R \bar{U}(i+j \mid i)\} \end{aligned}$$

H_∞ conditions

1) delay recovery and adjustment of volume targets: When $W(i) = 0$, the objective function (14) of the constraint (15) is minimal for all allowed uncertainties.

2) Robustness performance: With $X(i) = 0$ as the initial condition, the error states satisfy according to the system dynamics:

$$\begin{aligned} &\sum_{j=0}^{i_f-i} \left(X(i+j \mid i)^T X(i+j \mid i)\right)^{1/2} \leq \\ &\Gamma \sum_{j=0}^{i_f-i} \left(W(i+j \mid i)^T W(i+j \mid i)\right)^{1/2}. \end{aligned} \tag{16}$$